

Algebraic geometry and formal geometry

SGA 2, Exposé IX

English draft

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About this draft

This is an unofficial English translation of Exposé IX, “Algebraic geometry and formal geometry”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ IX

Algebraic geometry and formal geometry

The aim of this exposé is to generalize to the case of a morphism which is not proper the theorems 3.4.2 and 4.1.5 of EGA III.

1. The comparison theorem

Let $f: X \rightarrow X'$ be a morphism of preschemes, *separated* and of *finite type*. Suppose that X' is locally noetherian. Let Y' be a closed subset of X' and let $Y = f^{-1}(Y')$. Let $\widehat{X}$ and $\widehat{X'}$ be the formal completions of X and of X' along Y and along Y' . Let $\widehat{f}$ be the morphism deduced from f by passage to the completions.

$$(1) \quad \begin{array}{ccc} X & \longleftarrow & Y \\ f \downarrow & & \downarrow \\ X' & \longleftarrow & Y' \end{array}, \quad \begin{array}{ccc} \widehat{X} & \xrightarrow{j} & X \\ \widehat{f} \downarrow & & \downarrow f \\ \widehat{X'} & \xrightarrow{i} & X' \end{array}$$

One denotes by j (resp. i) the homomorphism from $\widehat{X}$ to X (resp. from $\widehat{X'}$ to X'). One knows that i and j are *flat*.

Let $\mathcal{I}'$ be an ideal of definition of Y' and let $\mathcal{J} = f^*(\mathcal{I}').\mathcal{O}_X$, it is an ideal of definition of Y . One thus has:

$$(2) \quad \widehat{X'} = (Y', \varprojlim_{k \in \mathbf{N}} \mathcal{O}_{X'}/\mathcal{I}'^{k+1}), \quad \widehat{X} = (Y, \varprojlim_{k \in \mathbf{N}} \mathcal{O}_X/\mathcal{J}^{k+1}).$$

For every $k \in \mathbf{N}$, set:

$$(3) \quad Y'_k = (Y', \mathcal{O}_{X'}/\mathcal{I}'^{k+1}), \quad Y_k = (Y, \mathcal{O}_X/\mathcal{J}^{k+1}).$$

Let F be a *coherent* $\mathcal{O}_X$ -Module. For every $k \in \mathbf{N}$, one sets:

$$(4) \quad F_k = F/\mathcal{J}^{k+1}F, \quad \widehat{F} = j^*(F) \simeq \varprojlim F_k.$$

If one sets:

$$(5) \quad R^i f_*(F)^\wedge = \varprojlim_{k \in \mathbf{N}} (R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'_k}), \quad i \in \mathbf{Z},$$

one has a natural homomorphism:

$$(6) \quad r_i : i^*(R^i f_*(F)) \longrightarrow R^i f_*(F)^\wedge,$$

which is an *isomorphism* when $R^i f_*(F)$ is *coherent*.

As is explained in EGA III 4.1.1, one has a commutative diagram:

$$(7) \quad \begin{array}{ccc} i^*(R^i f_*(F)) & \xrightarrow{\rho_i} & R^i \widehat{f}_*(\widehat{F}) \\ \downarrow r_i & & \downarrow \psi_i \\ R^i f_*(F)^\wedge & \xrightarrow{\varphi_i} & \varprojlim_{k \in \mathbf{N}} R^i f_*(F_k). \end{array}$$

In *loc. cit.* one finds a commutative diagram, since one knows that $R^i f_*(F)$ is coherent, and one identifies the source and the target of (6). In our case, $R^i f_*(F)$ will be coherent only for certain values of i , for which one will study (7).

Consider the graded ring

$$(8) \quad \mathcal{S} = \bigoplus_{k \in \mathbf{N}} \mathcal{I}'^k,$$

and the graded $\mathcal{S}$ -Module:

$$(9) \quad \mathcal{H}^i = \bigoplus_{k \in \mathbf{N}} R^i f_*(\mathcal{J}^k F), \quad i \in \mathbf{Z},$$

whose structure of $\mathcal{S}$ -Module is defined as follows.

The sheaf $R^i f_*(\mathcal{J}^k F)$ is associated to the presheaf which, to every affine open U' of X' , associates:

$$(10) \quad H^i(f^{-1}(U'), \mathcal{J}^k F|_{f^{-1}(U')}).$$

Let therefore U' be an affine open of X' , set

$$U = f^{-1}(U'),$$

and let $x' \in \mathcal{I}'^m(U')$. Let x be the image of x' in $\mathcal{J}^m(U)$. The homothety of ratio x in $F|_U$ maps $\mathcal{J}^k F|_U$ into $\mathcal{J}^{k+m} F|_U$, whence, by functoriality, a morphism:

$$(11) \quad \mu_{x',k}^i(U') : H^i(U, \mathcal{J}^k F|_U) \longrightarrow H^i(U, \mathcal{J}^{k+m} F|_U),$$

defined for every $i \in \mathbf{Z}$ and every $k \in \mathbf{N}$, which gives, by passage to the associated sheaf, the structure of graded $\mathcal{S}$ -Module of $\mathcal{H}^i$.

THEOREM 1.1. *Let n be an integer. Suppose that the graded $\mathcal{S}$ -Module $\mathcal{H}^i$ is of finite type for $i = n - 1$ and $i = n$, then:*

- (0) r_n and r_{n-1} are isomorphisms and $R^i \widehat{f}_*(\widehat{F})$ is coherent for $i = n - 1$;
- (1) for $i = n - 1$, ρ_i , φ_i and ψ_i are topological isomorphisms (in particular the filtration defined on $R^{n-1} f_*(F)$ by the kernels of the homomorphisms

$$(12) \quad R^{n-1} f_*(F) \longrightarrow R^{n-1} f_*(F_k)$$

is $\mathcal{I}'$ -good);

- (2) for $i = n$, ρ_i , φ_i and ψ_i are monomorphisms, moreover the filtration on $R^n f_*(F)$ is $\mathcal{I}'$ -good and ψ_n is an isomorphism;
- (3) The projective system of the $R^i f_*(F_k)$ satisfies, for $i = n - 2, n - 1$, the uniform Mittag-Leffler condition, i.e. there exists a fixed integer $k \geq 0$ such that, for every $p \geq 0$ and every $p' \geq p + k$, one has:

$$\text{Im}[R^i f_*(F_{p'}) \longrightarrow R^i f_*(F_p)] = \text{Im}[R^i f_*(F_{p+k}) \longrightarrow R^i f_*(F_p)].$$

Proceeding as in EGA III 4.1.8, it is easy to reduce to the case where X' is the spectrum of a noetherian ring A . In this case, one knows that

$$(13) \quad R^i f_*(F) = \widetilde{H^i(X, F)} \quad (\text{cf. 10}).$$

Let I be the ideal of A such that $\widetilde{I} = \mathcal{I}'$ and let

$$(14) \quad S = \bigoplus_{k \in \mathbf{N}} I^k,$$

$$(15) \quad H^i = \bigoplus_{k \in \mathbf{N}} H^i(X, \mathcal{J}^k F), \quad i \in \mathbf{Z},$$

where H^i is equipped with the structure of graded S -module defined by 11, where one has taken $U = X'$.

The proof is modelled on that of EGA III 4.1.5; let us give a summary.

One works with φ_i and ψ_i , to which correspond homomorphisms of modules:

$$(16) \quad \begin{array}{ccc} & H^i(\widehat{X}, \widehat{F}) & \\ & \downarrow \psi_i & \\ H^i(X, F)^\wedge & \xrightarrow{\varphi_i} \varprojlim_k & H^i(X, F_k). \end{array}$$

(a) One assumes only that H^i is a graded S -module of finite type. One deduces from this that *the filtration defined on $H^i(X, F)$ by the modules:*

$$(17) \quad R_k^i = \ker(H^i(X, F) \longrightarrow H^i(X, F_k))$$

is I -good. For this one uses the cohomology exact sequence:

$$(18) \quad H^i(X, \mathcal{J}^{k+1}F) \longrightarrow H^i(X, F) \longrightarrow H^i(X, F_k),$$

which proves that *the graded S -module $\bigoplus_{k \in \mathbf{N}} R_k^i$ is a quotient of the graded sub- S -module*

$$\bigoplus_{k \in \mathbf{N}} H^i(X, \mathcal{J}^{k+1}F)$$

of H^i , hence *is of finite type*, since S is noetherian. Whence this first point.

Set:

$$(19) \quad M^i = H^i(X, F), \quad H_k^i = H^i(X, F_k).$$

One has a commutative diagram:

$$(20) \quad \begin{array}{ccc} H^i(X, F)^\wedge & \xrightarrow{s_i} \varprojlim_k (M^i/R_k^i) & \\ & \searrow \varphi_i & \downarrow t_i \\ & & \varprojlim_k H_k^i, \end{array}$$

in which s_i is an *isomorphism*; indeed the filtration of $H^i(X, F)$ is I -good. Moreover t_i is a *monomorphism*; indeed the functor $\varprojlim$ is left exact, and, for every $k \geq 0$, the natural morphism $M^i/R_k^i \rightarrow H_k^i$ is a monomorphism, by definition of R_k^i .

To study the surjectivity of t_i one introduces:

$$(21) \quad Q_k^i = \text{coker}(H^i(X, F) \longrightarrow H^i(X, F_k)),$$

whence a projective system of exact sequences:

$$(22) \quad 0 \longrightarrow R_k^i \longrightarrow M^i \longrightarrow H_k^i \longrightarrow Q_k^i \longrightarrow 0.$$

Using the cohomology exact sequence:

$$(23) \quad H^i(X, F) \longrightarrow H^i(X, F_k) \longrightarrow H^{i+1}(X, \mathcal{J}^{k+1}F),$$

one sees that *the graded S -module*

$$(24) \quad Q^i = \bigoplus_{k \in \mathbf{N}} Q_k^i$$

is a graded sub- S -module of H^{i+1} . Moreover, for every $k \geq 0$, one has:

$$(25) \quad I^{k+1}Q_k^i = 0$$

since Q_k^i is the image of H_k^i .

(b) One assumes only that H^{i+1} is of finite type and one is concerned with t_i (forgetting s_i). Since S is noetherian, Q^i is of finite type; as $I^{k+1}Q_k^i$ is zero, one finds that there exist an integer $r \geq 0$ and an integer $k_0 \geq 0$ such that

$$(26) \quad I^r Q_k^i = 0 \quad \text{for } k \geq k_0.$$

It follows that the *projective system* $(Q_k^i)_{k \in \mathbf{N}}$ is *essentially zero*, hence that the *projective system* $(H_k^i)_{k \in \mathbf{N}}$ satisfies the *uniform Mittag-Leffler condition*. From the exact sequence (22) one deduces the exact sequence

$$(27) \quad 0 \longrightarrow M^i/R_k^i \longrightarrow H_k^i \longrightarrow Q_k^i \longrightarrow 0,$$

whence the exact sequence:

$$(28) \quad 0 \longrightarrow \varprojlim_k M^i/R_k^i \xrightarrow{t_i} \varprojlim_k H_k^i \longrightarrow \varprojlim_k Q_k^i.$$

Now the projective system $(Q_k^i)_{k \in \mathbf{N}}$ is essentially zero, hence t_i is an *isomorphism*.

(c) Let us prove that, if H^i is of finite type, ψ_i is an isomorphism. It suffices to apply EGA 0_{III} 13.3.1 taking as a basis of open sets of X the affine open sets. This is legitimate; indeed, by (b), the projective system $(H_k^{i-1})_{k \in \mathbf{N}}$ satisfies the Mittag-Leffler condition.

The theorem follows formally from (a), (b) and (c). One will remark that in fact the proof uses, at each step, the finiteness of H^i only for a single value of i .

Let us give some examples where the hypothesis of theorem 1.1 is satisfied.

COROLLARY 1.2. *Suppose that $\mathcal{I}'$ is generated by a section t' of $\mathcal{O}_{X'}$, and denote by t the corresponding section of $\mathcal{O}_X$. Let F be a coherent $\mathcal{O}_X$ -module and let n be an integer.*

Suppose that:

- (i) *t is F -regular (i.e. the homothety of ratio t in F is a monomorphism).*
- (ii) *$R^i f_*(F)$ is coherent for $i = n - 1$ and $i = n$.*

Then the hypothesis of theorem 1.1 is satisfied.

Indeed, one remarks that multiplication by t^k defines an isomorphism $F \xrightarrow{\sim} \mathcal{J}^k F$ and one deduces from this that

$$(29) \quad \mathcal{H}^i \simeq R^i f_*(F) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X'}[T],$$

where T is an indeterminate. Whence the conclusion.

COROLLARY 1.3¹. *Suppose that $X' = \text{Spec}(A)$, where A is a noetherian ring, separated and complete for the I -adic topology. Suppose that the S -module H^i is of finite type for $i = n - 1$ and $i = n$. (cf. 14 and 15). Then the hypotheses of theorem 1.1 are satisfied and one finds a commutative diagram of isomorphisms:*

$$(30) \quad \begin{array}{ccc} H^i(X, F) & \xrightarrow{\rho'_i} & H^i(\widehat{X}, \widehat{F}) \\ & \searrow \varphi'_i \quad \swarrow \psi_i & \\ & \varprojlim_k H^i(X, F_k) & \end{array} \quad \text{for } i = n - 1.$$

¹**N.D.E.** in the same vein, see the article of Chow, (Chow W.-L., “Formal functions on homogeneous spaces”, *Invent. Math.* **86** (1986), no. 1, p. 115–130). The author proves the following result. Let X be an algebraic variety over a field, homogeneous under an algebraic group G , and let Z be a complete subvariety of X of dimension > 0 . One assumes that Z generates X in the following sense: given $p \in Z$, let Γ_p be the set of elements of G sending p into Z . One then says that Z generates if the group generated by the connected component of 1 of Γ_p is all of G . In this case, every formal rational function of X along Z is algebraic; to be compared with the results of Hironaka and Matsumura cited in the editor’s note (??) page ??). In the line of the techniques introduced by these authors, let us mention the very pretty algebraization result due to Gieseker (Gieseker D., “On two theorems of Griffiths about embeddings with ample normal bundle”, *Amer. J. Math.* **99** (1977), no. 6, p. 1137–1150, theorems 4.1 and 4.2). Let X be a connected projective variety of dimension > 0 locally a complete intersection (over an algebraically closed field). One assumes that one has two embeddings of X into smooth projective varieties Y, W . Then, if the formal completions of X in Y and W are equivalent, there exists a scheme U containing X (as a closed subscheme) which embeds into Y and W as an étale neighbourhood of X in Y and W . In other words, formally equivalent implies étale-equivalent. See also the article of Faltings (Faltings G., “Formale Geometrie und homogene Räume”, *Invent. Math.* **64** (1981), p. 123–165).

One simply notes that $H^i(X, F)$ is of finite type hence isomorphic to its completion. One obtains (30) by transcribing into the category of A -modules the diagram of Modules (7), and replacing the left-hand vertical arrow by $H^i(X, F)$.

PROPOSITION 1.4. *Let A be a noetherian ring. Let $t \in A$ and suppose that A is separated and complete for the (tA) -adic topology. Set:*

$$(31) \quad X' = \text{Spec}(A), \quad Y' = V(t), \quad I = (tA).$$

Let T be a closed subset of X' , set

$$(32) \quad X = X' - T, \quad Y = Y' \cap X = Y' - (Y' \cap T).$$

Let F be a coherent $\mathcal{O}_X$ -Module. Finally let

$$(33) \quad T' = \{x \in X' \mid \text{codim}(\{\bar{x}\} \cap T, \{\bar{x}\}) = 1\}.$$

Suppose that:

- a) t is F -regular,
- b) $\text{prof}_{T'}(F) \geq n + 1$,
- c) A is a quotient of a regular noetherian ring.

Then, in the diagram (30), the morphisms ρ'_i , φ'_i and ψ_i are isomorphisms for $i < n$ and monomorphisms for $i = n$. Moreover ψ_n is an isomorphism.

By virtue of 3 and 2, it suffices to prove that $R^i f_*(F)$ is coherent for $i \leq n$, which follows from the finiteness theorem ??.

In particular:

EXAMPLE 1.5. One will apply 4 when A is a local ring and t belongs to the radical $\mathfrak{r}(A)$ of A . One will then take $T = \{\mathfrak{r}(A)\}$. In this case, for $n = 1$ one finds the following statement:

If A noetherian is separated and complete for the t -adic topology and a quotient of a regular ring (for example if A is complete), if moreover t is F -regular and if $\text{prof } F_x \geq 2$ for every $x \in \text{Spec}(A)$ such that $\dim A/x = 1$, then the natural homomorphism

$$\Gamma(X, F) \longrightarrow \Gamma(\hat{X}, \hat{F})$$

is an isomorphism.

Indeed, retaining the notations of 4, one has $T = \{\mathfrak{r}(A)\}$ and formula (33) says that

$$T' = \{x \in \text{Spec}(A) \mid \dim A/x = 1\}.$$

2. Existence theorem

Let us first state EGA III 3.4.2 in a slightly more general form.

Let $f : \mathfrak{X} \rightarrow \mathfrak{X}'$ be an adic morphism² of formal preschemes, with $\mathfrak{X}'$ noetherian. Let $\mathcal{I}'$ be an ideal of definition of $\mathfrak{X}'$; since f is adic, $f^* \mathcal{I}' = \mathcal{J}$ is³ an ideal of definition of $\mathfrak{X}$.

For every $n \in \mathbf{N}$, set

$$(34) \quad \mathfrak{X}_n = (\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathcal{J}^{n+1}),$$

this is an ordinary prescheme which has the same underlying topological space as $\mathfrak{X}$.

Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\mathfrak{X}}$ -Module. For every $k \in \mathbf{N}$, the $\mathcal{O}_{\mathfrak{X}_k}$ -Modules

$$(35) \quad F_k = \mathfrak{F}/\mathcal{J}^{k+1}\mathfrak{F}$$

are coherent. For every i , one has a homomorphism

$$(36) \quad \psi_i : R^i f_*(\mathfrak{F}) \longrightarrow \varprojlim_k R^i f_*(F_k),$$

deduced by functoriality from the natural homomorphism:

$$(37) \quad \mathfrak{F} \longrightarrow F_k.$$

²This hypothesis is not essential, cf. ??, p. ??.

³N.D.E. by the very definition, cf. EGA I 10.12.1.

Set

$$(38) \quad \mathbf{S} = \mathrm{gr}_{\mathcal{I}'} \mathcal{O}_{\mathfrak{X}'} = \bigoplus_{k \in \mathbf{N}} \mathcal{I}'^k / \mathcal{I}'^{k+1},$$

$$(39) \quad \mathrm{gr}_{\mathcal{J}}(\mathfrak{F}) = \bigoplus_{k \in \mathbf{N}} \mathcal{J}^k \mathfrak{F} / \mathcal{J}^{k+1} \mathfrak{F},$$

$$(40) \quad \mathbf{K}^i = R^i f_*(\mathrm{gr}_{\mathcal{J}}(\mathfrak{F})) = \bigoplus_{k \in \mathbf{N}} R^i f_*(\mathcal{J}^k \mathfrak{F} / \mathcal{J}^{k+1} \mathfrak{F}).$$

It is clear that $\mathbf{K}^i$ is equipped with a structure of graded $\mathbf{S}$ -Module.

THEOREM 2.1. *Suppose that $\mathbf{K}^i$ is a graded $\mathbf{S}$ -Module of finite type for $i = n - 1$, $i = n$, $i = n + 1$, then:*

- (i) $R^n f_*(\mathfrak{F})$ is coherent.
- (ii) The homomorphism ψ_n (36) is a topological isomorphism. The natural filtration of the second member of (36) is $\mathcal{I}'$ -good.
- (iii) The projective system of the $R^n f_*(F_k)$ satisfies the uniform Mittag-Leffler condition.

The proof is very easy from EGA 0_{III}13.7.7 (cf. EGA III 3.4.2), provided one corrects the text on page 78 as indicated in (EGA III 2, Err_{III}24).

THEOREM 2.2⁴. *Let A be a noetherian adic ring and let I be an ideal of definition of A . Let T be a closed subset of $X' = \mathrm{Spec}(A)$. Suppose that I is generated by some $t \in A$. Let us resume the notations 31, 32 and 33. Let $\mathfrak{F}$ be a coherent $\mathcal{O}_{\widehat{X}}$ -Module. Set*

$$(8) \quad F_0 = \mathfrak{F} / \mathcal{J} \mathfrak{F},$$

where $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$ is an ideal of definition of $\widehat{X}$. Suppose that A is a quotient of a regular noetherian ring and that:

- (1) t is $\mathfrak{F}$ -regular,
- (2) $\mathrm{prof}_T F_0 \geq 2$.

Then there exists a coherent $\mathcal{O}_X$ -Module F such that $\widehat{F} \simeq \mathfrak{F}$.

It suffices to prove that $\widehat{f}_*(\mathfrak{F})$ is a coherent $\mathcal{O}_{\widehat{X}}$ -Module, where $\widehat{f} : \widehat{X} \rightarrow \widehat{X}'$ is the morphism of formal preschemes deduced from the injection of X into X' by completion with respect to t . Indeed, A is separated and complete for the t -adic topology, there will therefore exist an A -module F' whose completion will be isomorphic to $\widehat{f}_*(\mathfrak{F})$. Since X is an open subset of X' , one may take $F = \widetilde{F'}|_X$.

It remains to show that 2.1 is applicable to the morphism of formal preschemes $\widehat{f}$ and to $\mathfrak{F}$. Now, by hypothesis (1), for every $k \in \mathbf{N}$ one has an isomorphism:

$$\mathcal{J}^k \mathfrak{F} / \mathcal{J}^{k+1} \mathfrak{F} \longrightarrow \mathfrak{F} / \mathcal{J} \mathfrak{F},$$

whence it follows that the hypothesis of 2.1 will be satisfied if one knows that

$$R^i f_*(F_0) \text{ is coherent for } i \leq 1.$$

Now this follows from (2) and from the finiteness theorem ???. Whence the conclusion.

It remains to specialize this statement by assuming that A is a local ring.

⁴**N.D.E.** many algebraization statements have been obtained since, not to mention those cited below, cf. the articles of Faltings or of Mme Raynaud cited in the editor's note (??) p. ?? and (??) p. ?? respectively. One thinks in particular of the results of Artin (see in particular Artin M., "Algebraization of formal moduli. I", in *Global Analysis (Papers in Honor of K. Kodaira)*, Univ. Tokyo Press, Tokyo, 1969, p. 21–71), but also of the recent results on algebraicity of leaves of foliations; see in particular Bost J.-B., "Algebraic leaves of algebraic foliations over number fields", *Publ. Math. Inst. Hautes Études Sci.* **93** (2001), p. 161–221 and Chambert-Loir A. "Algebraicity theorems in Diophantine geometry (after J.-B. Bost, Y. André, D. & G. Chudnovsky)", in *Séminaire Bourbaki, Vol. 2000/2001*, Astérisque, vol. 282, Société mathématique de France, Paris, 2002, Exp. 886, p. 175–209 and the references cited. In particular, one will find in these two articles discussions of the link between algebraization questions and the theory of Diophantine approximation.

COROLLARY 2.3. *Let A be a noetherian local ring and let $t \in \mathfrak{r}(A)$ be an element of the radical of A . Suppose that A is separated and complete for the t -adic topology, and, moreover, a quotient of a regular ring (for example, suppose that A is a complete noetherian local ring). Set*

$$(9) \quad X' = \operatorname{Spec} A, \quad T = \{\mathfrak{r}(A)\},$$

and let us resume the notations (31), (32) and (33). Let $\mathfrak{F}$ be an $\mathcal{O}_{\widehat{X}}$ -Module. Suppose that:

- (1) *t is $\mathfrak{F}$ -regular,*
- (2) *$\operatorname{prof}_{T'} F_0 \geq 2$, with $F_0 = \mathfrak{F}/\mathcal{J}\mathfrak{F}$ and $\mathcal{J} = t\mathcal{O}_{\widehat{X}}$.*

Then there exists a coherent $\mathcal{O}_X$ -Module F such that $\widehat{F} \simeq \mathfrak{F}$.

Let us remark that here T' is the set of prime ideals $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$.